

Airbnb Dynamic Pricing Recommendation Engine

From historical listing data to market-based pricing guidance

Project Objective

Analyze historical Airbnb data to compare listing prices with comparable market benchmarks and generate transparent pricing recommendations.

Dataset & Tools

48,895 listings · 5 neighbourhood groups · 3 room types

PYTHON

EXCEL

POWER BI

01 · BUSINESS PROBLEM

Why Data-Driven Pricing Matters

Hosts need to know whether their listing is underpriced, competitively priced, or overpriced relative to similar properties. Airbnb prices vary considerably by location and room type, so one fixed pricing strategy is not appropriate for every listing.

❑ **Core business question:** What price should a host consider charging based on comparable listings in the same market?

02 · ANALYTICS WORKFLOW

End-to-End Project Pipeline



Python

EDA · Cleaning · Feature Engineering · Recommendation Logic



Excel

Engineered Dataset · Connected Source · Pricing Fields



Power BI

Interactive Dashboard · KPIs · Market Insights · Recommendations

Dataset dimensions used in the project: listing identity, host information, location, coordinates, room type, price, minimum nights, reviews, review frequency, host listing count and availability.

03 · PYTHON EDA

Exploratory Data Analysis — Actual Notebook Outputs



Average Price by Borough

Manhattan leads at ~\$195, followed by Brooklyn (~\$125), Staten Island (~\$115), Queens (~\$100), and Bronx (~\$85).

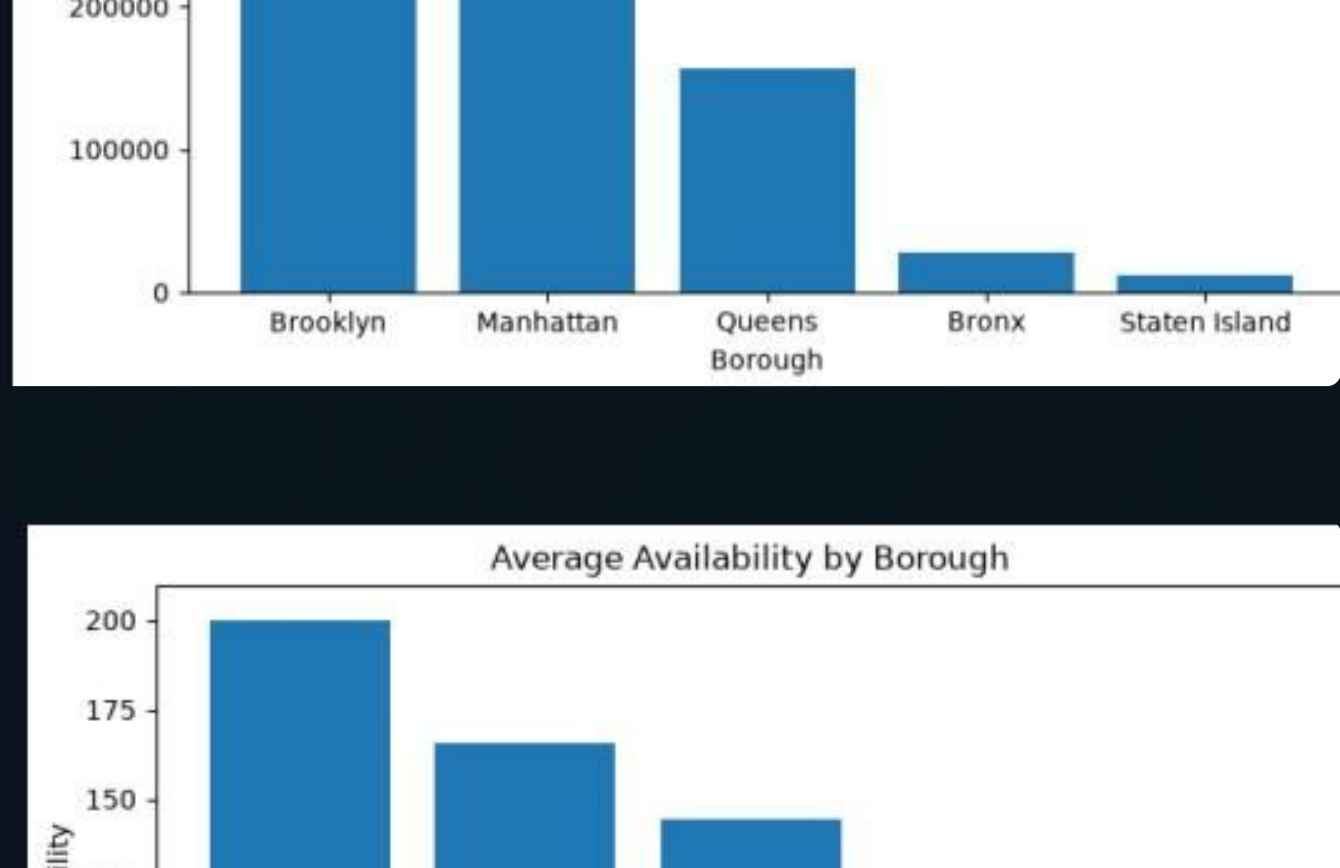


Total Listings by Borough

Manhattan (~22,000) and Brooklyn (~20,000) dominate supply, while Queens (~5,500), Bronx (~1,000), and Staten Island (~500) represent smaller markets.

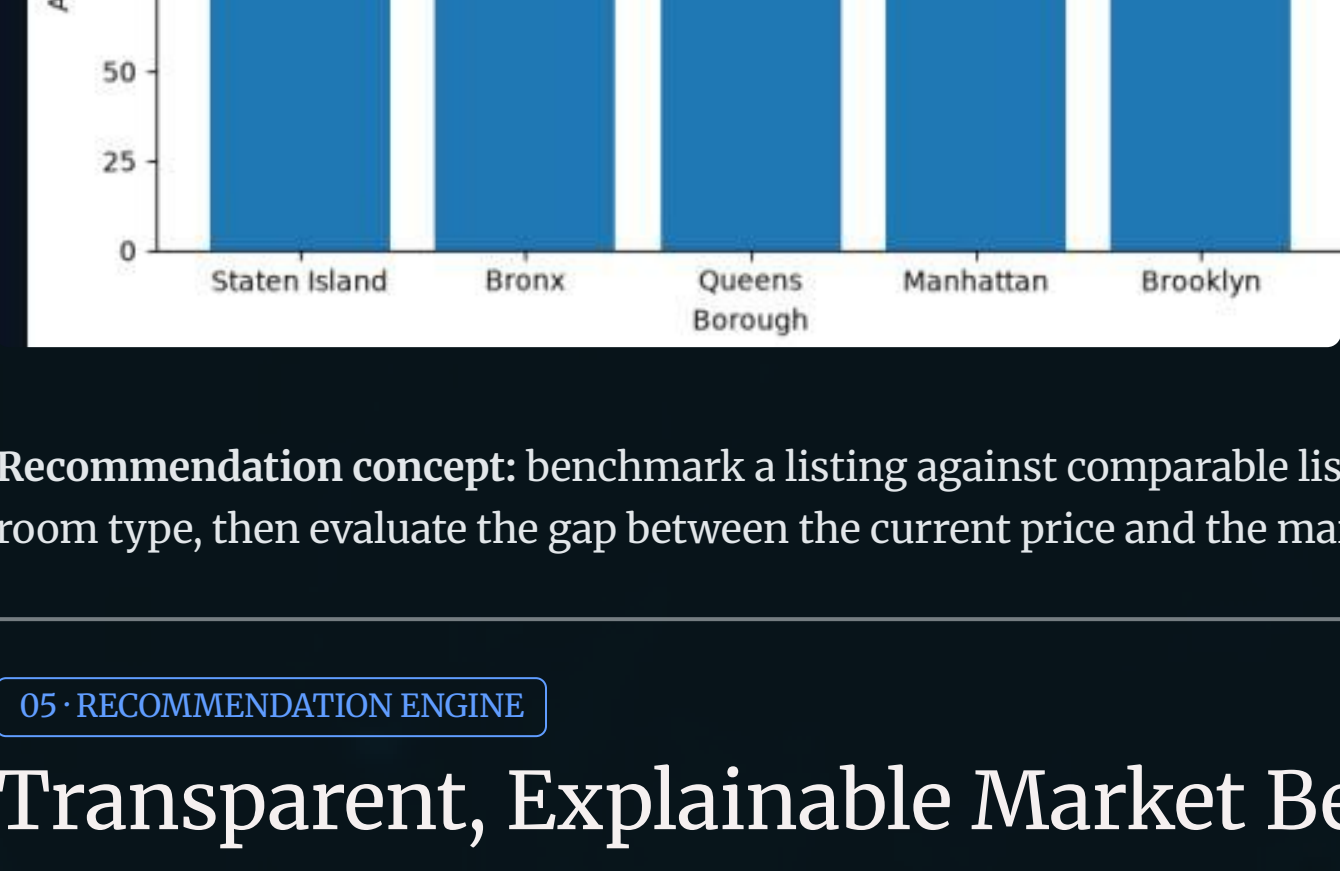
04 · FEATURE ENGINEERING & PRICING LOGIC

Turning Raw Data into Decision-Ready Features



Reviews by Borough

Brooklyn (~480K) and Manhattan (~450K) lead in review volume, followed by Queens (~150K), Bronx (~30K), and Staten Island (~10K).



Average Availability by Borough

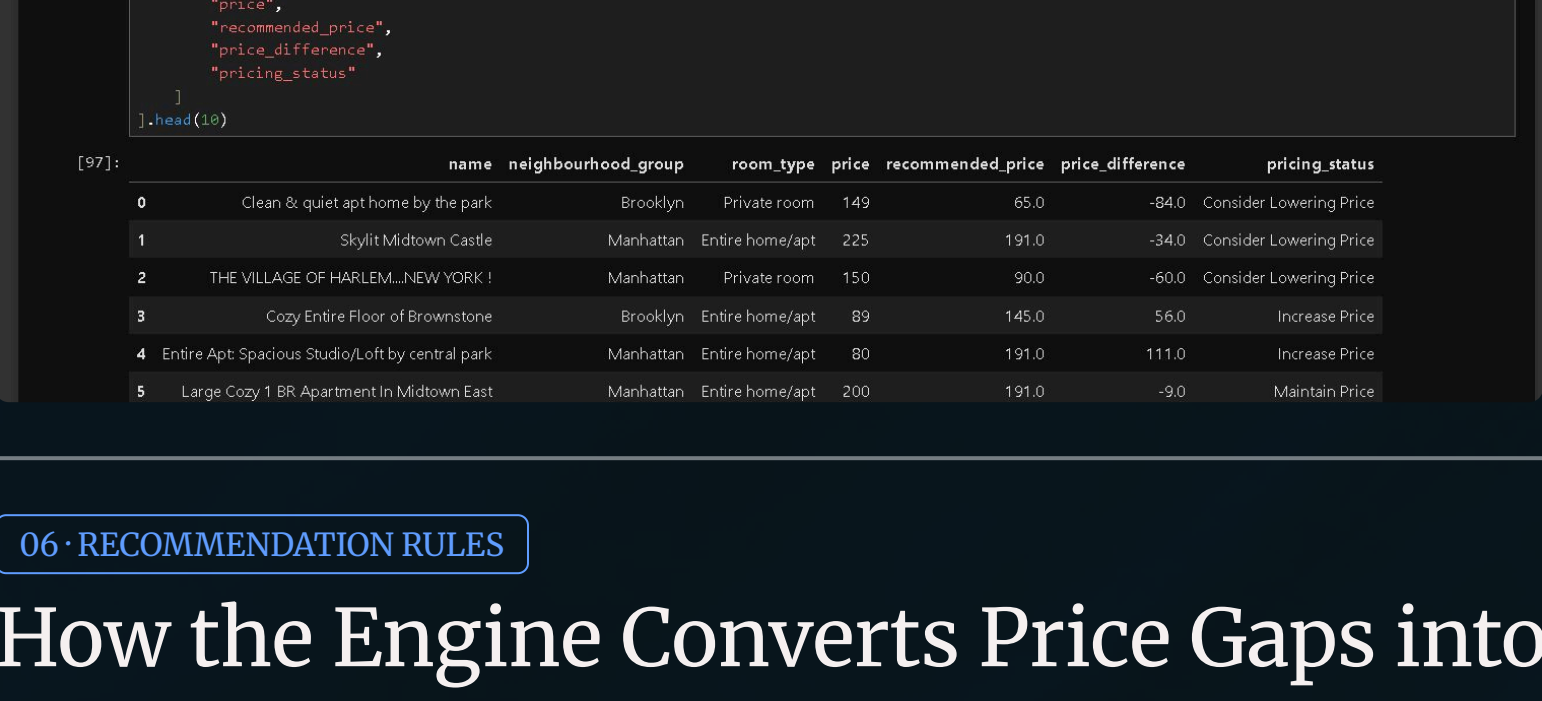
Staten Island (~200) and Bronx (~165) show the highest availability, while Brooklyn (~100) and Manhattan (~110) have the lowest.

Recommendation concept: benchmark a listing against comparable listings using neighbourhood group and room type, then evaluate the gap between the current price and the market benchmark.

05 · RECOMMENDATION ENGINE

Transparent, Explainable Market Benchmarking

Recommended Price = Median price of comparable listings (neighbourhood group + room type). **Price Difference** = **Recommended Price** – **Current Price**. The recommendation is intended as a market-based pricing guide, not a guaranteed optimal-price prediction.



06 · RECOMMENDATION RULES

How the Engine Converts Price Gaps into Actions

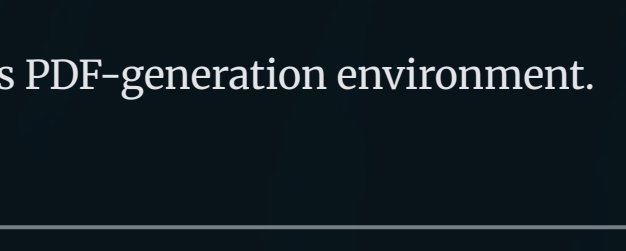
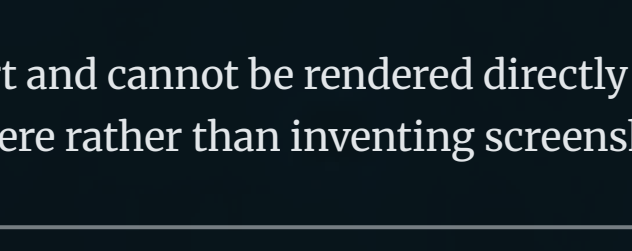
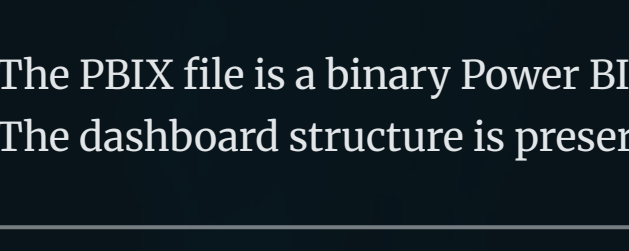
Price Difference	Recommendation	Business Interpretation
> 10% below benchmark	INCREASE PRICE	Potential underpricing opportunity
Within ±10%	MAINTAIN PRICE	Price is broadly aligned with comparable listings
> 10% above benchmark	CONSIDER LOWERING PRICE	Potential overpricing relative to comparable listings

The engineered Excel dataset contains the pricing-status output used by the Power BI recommendation page.

07 · POWER BI DASHBOARD

Interactive Decision Support for Airbnb Pricing

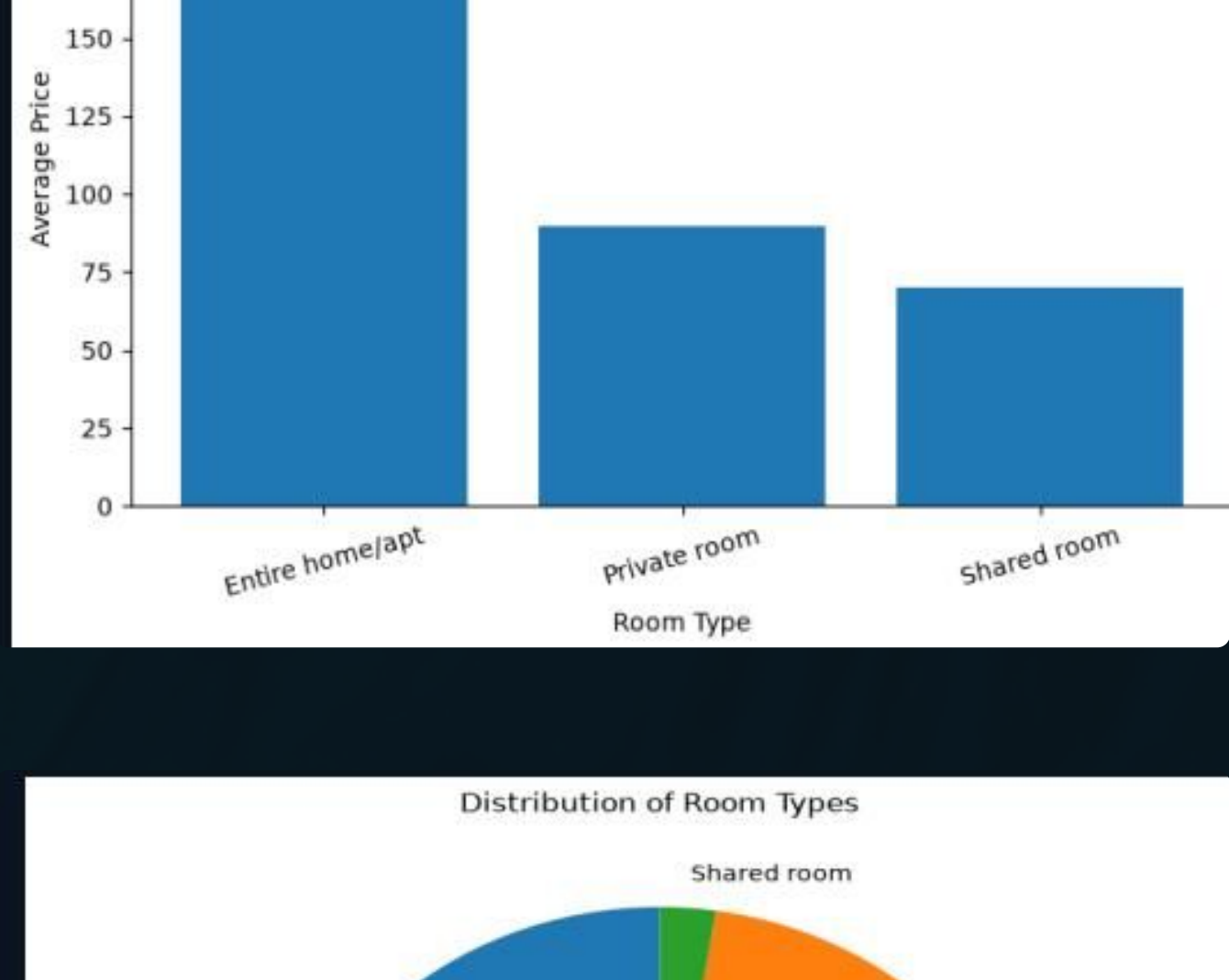
The Power BI report is the presentation layer of the project: it allows users to explore the Airbnb market, compare pricing patterns, and interact with the recommendation outputs.



The PBIX file is a binary Power BI report and cannot be rendered directly in this PDF-generation environment. The dashboard structure is preserved here rather than inventing screenshots.

08 · VISUAL EVIDENCE

Actual Notebook Outputs Supporting the Recommendation Engine



Average Price by Room Type

Entire home/apt (~\$210) commands the highest average price, followed by Private room (~\$90) and Shared room (~\$70).



Distribution of Room Types

Entire home/apt accounts for 52.0% of listings, Private room for 45.7%, and Shared room for just 2.4%.

These visuals are taken directly from the executed Python notebook and provide visual evidence of the analytical work behind the recommendation engine.

09 · BUSINESS INSIGHTS & CONCLUSION

From Descriptive Analytics to Prescriptive Pricing Guidance

Key Dimensions Location and room type are important dimensions for comparing Airbnb prices.	Transparent Benchmark A market benchmark can identify listings that may be underpriced or overpriced relative to comparable listings. The recommendation framework is transparent and explainable because the benchmark is based on comparable listings rather than an opaque black-box model.	Interactive Analysis Power BI makes the analysis interactive and allows users to move from market-level patterns to actionable pricing recommendations.
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⚠ **Limitation:** The dataset does not contain actual bookings, occupancy, revenue, or real-time demand. Therefore, the output should be treated as market-based pricing guidance rather than an exact optimal-price forecast.

Final takeaway: This project demonstrates a complete analytics workflow using Python, Excel, and Power BI to convert historical Airbnb listing data into a practical pricing recommendation engine.